

INTEGER SOLUTIONS OF $Y^2 + X^2Y + XZ^2 + 1 = 0$

ABSTRACT. We give a complete and effective classification of the integer solutions of

$$y^2 + x^2y + xz^2 + 1 = 0.$$

For positive x , every solution is obtained from a finite quadratic-residue and square test. For negative square x , the equation splits into a finite divisor problem, whereas for negative nonsquare x it becomes a generalized Pell equation whose admissible solutions are partitioned into finitely many explicitly bounded orbits. In particular, each fixed negative nonsquare value of x supports either no solutions or infinitely many, and an explicit infinite orbit with $x = -145$ is exhibited.

1. STATEMENT OF THE CLASSIFICATION

We study the Diophantine equation

$$(1) \quad y^2 + x^2y + xz^2 + 1 = 0, \quad (x, y, z) \in \mathbb{Z}^3.$$

For $n \geq 2$, define

$$(2) \quad \mathcal{R}_n = \{r \in \{1, \dots, n-1\} : r^2 \equiv -1 \pmod{n}\},$$

and, for $r \in \mathcal{R}_n$ and $0 \leq j \leq n-1$, define

$$(3) \quad H_{n,r,j} = jn(n-j) + r(n-2j) - \frac{r^2+1}{n}.$$

The last expression is an integer because $n \mid r^2 + 1$.

For the nonsquare negative case, put

$$(4) \quad D_n = n^4 - 4.$$

When $n \geq 2$ is nonsquare, choose any integers $p_n, q_n > 0$ satisfying

$$(5) \quad p_n^2 - nq_n^2 = 1,$$

and write

$$(6) \quad \lambda_n = (p_n + q_n\sqrt{n})^6 = P_n + Q_n\sqrt{n}.$$

We shall prove that such p_n, q_n always exist and that

$$(7) \quad P_n \equiv 1 \pmod{2}, \quad Q_n \equiv 0 \pmod{2}.$$

Define the reduced seed set

$$(8) \quad \mathcal{C}_n = \left\{ (A, B) \in \mathbb{Z}^2 : A^2 - nB^2 = D_n, \right. \\ \left. A \equiv n^2 \pmod{2}, \quad B \equiv 0 \pmod{2}, \right. \\ \left. \sqrt{\frac{D_n}{\lambda_n}} \leq A + B\sqrt{n} < \sqrt{D_n\lambda_n} \right\}.$$

Here and below, expressions involving $\sqrt{n}$ are compared in the usual real embedding.

Theorem 1.1 (Complete classification). *Every integer solution of (1) occurs in exactly one of the following three cases.*

(I) Positive x . For $n \geq 2$, all solutions with $x = n$ are

$$(9) \quad (x, y, z) = (n, -(r + jn), \varepsilon s),$$

where

$$r \in \mathcal{R}_n, \quad 0 \leq j \leq n-1, \quad \varepsilon \in \{\pm 1\}, \quad s \in \mathbb{Z}_{>0},$$

and

$$(10) \quad s^2 = H_{n,r,j}.$$

Conversely, every choice satisfying these conditions gives a solution.

(II) Negative square x . For $a \geq 1$, all solutions with $x = -a^2$ are

$$(11) \quad (x, y, z) = \left(-a^2, \frac{A + B - 2a^4}{4}, \frac{B - A}{4a} \right),$$

where $A, B \in \mathbb{Z}$ satisfy

$$(12) \quad AB = a^8 - 4, \quad 4a \mid B - A, \quad 4 \mid A + B - 2a^4.$$

Conversely, every factor pair satisfying (12) gives a solution.

(III) Negative nonsquare x . Let $n \geq 2$ be nonsquare. For every

$$(A, B) \in \mathcal{C}_n, \quad m \in \mathbb{Z}, \quad \varepsilon \in \{\pm 1\},$$

write uniquely

$$(13) \quad U + V\sqrt{n} = \varepsilon(A + B\sqrt{n})\lambda_n^m, \quad U, V \in \mathbb{Z}.$$

Then

$$(14) \quad (x, y, z) = \left(-n, \frac{U - n^2}{2}, \frac{V}{2} \right)$$

solves (1); and every solution with $x = -n$ arises uniquely in this fashion, once the chosen unit λ_n is fixed.

The set $\mathcal{C}_n$ is finite and can be found by a bounded search. Consequently, for each fixed nonsquare $n \geq 2$, the fiber $x = -n$ is either empty or infinite.

The classification is effective in the literal sense. Case (I) requires finitely many tests of whether the integers in (3) are positive squares; Case (II) requires only the finitely many divisors of $a^8 - 4$; and Case (III) requires a Pell unit and a search inside the explicit bounds proved in proposition 4.3 below.

2. ELEMENTARY REDUCTIONS AND CONGRUENCE OBSTRUCTIONS

Lemma 2.1 (Symmetries and degenerate cases). *Equation (1) is invariant under*

$$(15) \quad (x, y, z) \mapsto (x, y, -z), \quad (x, y, z) \mapsto (x, -x^2 - y, z).$$

Moreover, every integer solution satisfies

$$(16) \quad x \neq 0, \quad z \neq 0.$$

Proof. The first map in (15) leaves z^2 unchanged. For the second map, if $y' = -x^2 - y$, then

$$(y')^2 + x^2 y' = (-x^2 - y)^2 + x^2(-x^2 - y) = y^2 + x^2 y.$$

Thus both maps preserve the left-hand side of (1).

If $x = 0$, then (1) becomes $y^2 + 1 = 0$, which has no integer solution. If $z = 0$, then

$$y(y + x^2) = -1.$$

The only ordered pairs of integers with product -1 are $(1, -1)$ and $(-1, 1)$. In the first case $y = 1$ and $y + x^2 = -1$, giving $x^2 = -2$; in the second, $y = -1$ and $y + x^2 = 1$, giving $x^2 = 2$. Both are impossible in integers. Hence (16) holds. $\square$

The basic completed-square transformation is the source of all three cases in theorem 1.1.

Lemma 2.2 (Norm-form reduction). *Let (x, y, z) be an integer solution and write*

$$x = \delta n, \quad n = |x| \geq 1, \quad \delta \in \{1, -1\}.$$

Set

$$(17) \quad U = 2y + n^2, \quad V = 2z.$$

Then

$$(18) \quad U^2 + \delta n V^2 = n^4 - 4,$$

with the parity conditions

$$(19) \quad U \equiv n^2 \pmod{2}, \quad V \equiv 0 \pmod{2}.$$

Conversely, every pair $(U, V) \in \mathbb{Z}^2$ satisfying (18) and (19) gives an integer solution through

$$(20) \quad y = \frac{U - n^2}{2}, \quad z = \frac{V}{2}.$$

Thus (17) is a bijection between the solutions with fixed $x = \delta n$ and the parity-admissible solutions of (18).

Proof. Multiplying (1) by 4 and completing the square in y gives

$$(2y + x^2)^2 + 4xz^2 = x^4 - 4.$$

Since $x^2 = n^2$ and $x^4 = n^4$, substitution of (17) yields (18). The congruences in (19) follow immediately from the definitions.

Conversely, (19) makes the quantities in (20) integral. Reversing the completed-square calculation then gives (1). The two constructions are visibly inverse to each other. $\square$

A second elementary restriction eliminates most possible values of $|x|$ before any further analysis is needed.

Proposition 2.3 (The -1 residue obstruction). *If (x, y, z) satisfies (1) and $n = |x|$, then*

$$(21) \quad y^2 \equiv -1 \pmod{n}.$$

For a positive integer n , the congruence $t^2 \equiv -1 \pmod{n}$ is solvable if and only if

$$(22) \quad n = 2^\delta \prod_{i=1}^k p_i^{e_i}, \quad \delta \in \{0, 1\}, \quad p_i \equiv 1 \pmod{4},$$

where the p_i are distinct odd primes and the e_i are positive integers. In that event, the number of residue classes modulo n satisfying $t^2 \equiv -1 \pmod{n}$ is 2^k .

Proof. Reducing (1) modulo $n = |x|$ annihilates both terms containing x , and gives (21).

We now classify the moduli for which the latter congruence is solvable. First suppose $t^2 \equiv -1 \pmod{n}$. No square is congruent to $-1 \equiv 3 \pmod{4}$, so $4 \nmid n$. If an odd prime p divides n , then $t^2 \equiv -1 \pmod{p}$. In the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$, the residue class of t has order exactly 4: its square is -1 , whereas neither t nor t^2 is 1. The order of an element divides the order $p-1$ of the finite group, so $4 \mid p-1$, or equivalently $p \equiv 1 \pmod{4}$. This proves the necessity of (22).

For the converse, let $p \equiv 1 \pmod{4}$ be prime. We first recall the short proof of Wilson's congruence. In $(\mathbb{Z}/p\mathbb{Z})^\times$, every element other than 1 and -1 can be paired with its distinct inverse; hence the product of all nonzero residue classes is -1 . Therefore

$$(p-1)! \equiv -1 \pmod{p}.$$

On the other hand,

$$(p-1)! = \prod_{a=1}^{(p-1)/2} a(p-a) \equiv (-1)^{(p-1)/2} \left(\left(\frac{p-1}{2} \right)! \right)^2 \pmod{p}.$$

Because $p \equiv 1 \pmod{4}$, the exponent $(p-1)/2$ is even, and therefore

$$\left(\left(\frac{p-1}{2}\right)!\right)^2 \equiv -1 \pmod{p}.$$

Thus -1 has a square root modulo p .

Such a root lifts to every prime power p^e . Indeed, suppose

$$r_e^2 + 1 = c_e p^e$$

for some integer c_e . Since $p \nmid r_e$, there is a unique $h \pmod{p}$ satisfying

$$c_e + 2r_e h \equiv 0 \pmod{p}.$$

Then

$$(r_e + hp^e)^2 + 1 \equiv r_e^2 + 1 + 2r_e hp^e \equiv 0 \pmod{p^{e+1}},$$

so the root lifts inductively. The two roots modulo p lift to two roots modulo each p^e . There are no others: if $r^2 \equiv s^2 \equiv -1 \pmod{p^e}$, then

$$p^e \mid (r-s)(r+s).$$

The two factors cannot both be divisible by p , because that would imply $p \mid 2r$ and hence $p \mid r$, contrary to $r^2 \equiv -1 \pmod{p}$. Thus one factor is prime to p and the other is divisible by p^e , so $r \equiv \pm s \pmod{p^e}$.

Modulo 2 there is exactly one root, namely 1. The Chinese remainder theorem now combines independently chosen roots modulo the pairwise coprime prime-power factors in (22). This proves existence and gives exactly two choices for each odd prime factor and one choice for the factor 2, hence 2^k roots in total. $\square$

Remark 2.4. By proposition 2.3, every solution of (1) has

$$|x| = 2^\delta \prod_{i=1}^k p_i^{e_i} \quad \text{with} \quad \delta \in \{0, 1\}, \quad p_i \equiv 1 \pmod{4}.$$

This is only a necessary condition for the original equation; the subsequent square, divisor, or Pell conditions decide whether a given candidate actually occurs.

3. THE POSITIVE AND NEGATIVE-SQUARE CASES

We first treat $x > 0$. Here the norm equation in lemma 2.2 is positive definite, and a direct argument gives the sharper finite parametrization in Theorem 1.1(I).

Proposition 3.1 (Positive x). *The solutions with $x > 0$ are exactly those in (9)–(10).*

Proof. Write $x = n > 0$. Equation (1) becomes

$$(23) \quad y(y + n^2) = -(nz^2 + 1) < 0.$$

Therefore

$$-n^2 < y < 0.$$

In particular, $n = 1$ is impossible, because there is no integer strictly between -1 and 0 . Set

$$a = -y.$$

Then

$$(24) \quad 1 \leq a \leq n^2 - 1,$$

and (23) is equivalent to

$$(25) \quad a(n^2 - a) = nz^2 + 1.$$

Reducing modulo n gives

$$(26) \quad a^2 \equiv -1 \pmod{n}.$$

Let r be the least positive residue of a modulo n . Equation (26) implies $r \neq 0$, so $r \in \mathcal{R}_n$. By (24), there is a unique integer j with

$$(27) \quad a = r + jn, \quad 0 \leq j \leq n-1.$$

Substituting (27) into (25) and dividing by n gives

$$\begin{aligned} z^2 &= \frac{(r + jn)(n^2 - r - jn) - 1}{n} \\ &= jn(n - j) + r(n - 2j) - \frac{r^2 + 1}{n} \\ &= H_{n,r,j}. \end{aligned}$$

By lemma 2.1, $z \neq 0$, so $H_{n,r,j}$ must be a positive square, say s^2 with $s > 0$, and $z = \varepsilon s$ for a unique $\varepsilon \in \{\pm 1\}$. Since $y = -a = -(r + jn)$, every solution has the asserted form.

Conversely, suppose n, r, j, s, ε satisfy the conditions in Theorem 1.1(I), and put $a = r + jn$. The identity $s^2 = H_{n,r,j}$ is exactly

$$a(n^2 - a) = ns^2 + 1.$$

Taking $x = n$, $y = -a$, and $z = \varepsilon s$ therefore gives

$$y^2 + x^2y + xz^2 + 1 = a^2 - n^2a + ns^2 + 1 = 0.$$

Thus every listed triple is a solution. $\square$

Example 3.2. For $n = 2$, one has $\mathcal{R}_2 = \{1\}$, and the two admissible values $j = 0, 1$ both give $H_{2,1,j} = 1$. Thus

$$(2, -1, \pm 1), \quad (2, -3, \pm 1)$$

are exactly the solutions with $x = 2$.

For $n = 5$, one has $\mathcal{R}_5 = \{2, 3\}$. The positive-square tests yield exactly

$$(5, -2, \pm 3), \quad (5, -7, \pm 5), \quad (5, -18, \pm 5), \quad (5, -23, \pm 3).$$

The pairing of the y -coordinates is the involution $y \mapsto -25 - y$ from lemma 2.1.

We next treat negative square values of x . In this case the indefinite norm form splits over $\mathbb{Z}$ itself.

Proposition 3.3 (Negative square x). *The solutions with $x = -a^2$, $a \geq 1$, are exactly those in (11)–(12). For each fixed a , there are only finitely many such solutions.*

Proof. Set $x = -a^2$. In lemma 2.2, $n = a^2$ and $\delta = -1$, so

$$(28) \quad U^2 - a^2V^2 = a^8 - 4, \quad U \equiv a^4 \pmod{2}, \quad V \equiv 0 \pmod{2}.$$

Factor the left-hand side and put

$$(29) \quad A = U - aV, \quad B = U + aV.$$

Then

$$(30) \quad AB = a^8 - 4.$$

Moreover,

$$B - A = 2aV = 4az$$

and

$$A + B - 2a^4 = 2U - 2a^4 = 4y.$$

Thus every solution determines a factor pair satisfying all three conditions in (12), and solving (29) for y, z gives precisely (11).

Conversely, suppose A, B satisfy (12), and define x, y, z by (11). The divisibility assumptions make y and z integers. Put

$$U = 2y + a^4 = \frac{A + B}{2}, \quad V = 2z = \frac{B - A}{2a}.$$

Then

$$U - aV = A, \quad U + aV = B,$$

so

$$U^2 - a^2V^2 = AB = a^8 - 4.$$

The inverse direction of lemma 2.2 now shows that (x, y, z) solves (1).

Finally, $a^8 - 4 \neq 0$ for every integer $a \geq 1$. Hence it has only finitely many integer divisors, and therefore only finitely many ordered factor pairs (A, B) . This proves the finiteness assertion. $\square$

Example 3.4. When $a = 1$, one has $AB = -3$. The divisibility conditions in (12) select exactly the four ordered factor pairs that produce

$$(-1, 0, \pm 1), \quad (-1, -1, \pm 1).$$

These are therefore all solutions with $x = -1$.

4. THE NONSQUARE PELL CASE

We now prove the machinery required for Theorem 1.1(III). The first lemma supplies a positive unit of norm 1 without appealing to continued fractions.

Lemma 4.1 (Elementary existence of a Pell unit). *Let $n \geq 2$ be nonsquare. Then there exist positive integers p, q such that*

$$(31) \quad p^2 - nq^2 = 1.$$

Equivalently, $\mathbb{Z}[\sqrt{n}]$ contains a real unit $p + q\sqrt{n} > 1$ of norm 1.

Proof. For each integer $N \geq 1$, partition $[0, 1)$ into N half-open intervals of length $1/N$ and consider the $N + 1$ fractional parts

$$\{j\sqrt{n}\}, \quad 0 \leq j \leq N.$$

Two lie in the same interval. Subtracting them gives integers p_N, q_N with

$$(32) \quad 1 \leq q_N \leq N, \quad |q_N\sqrt{n} - p_N| < \frac{1}{N}.$$

There are infinitely many distinct pairs (p_N, q_N) satisfying such inequalities as N varies. Indeed, if only finitely many occurred, then, because $\sqrt{n}$ is irrational, the positive numbers $|q\sqrt{n} - p|$ attached to those pairs would have a positive minimum; choosing N larger than its reciprocal would contradict (32).

For every such pair, put

$$k_N = p_N^2 - nq_N^2.$$

This is a nonzero integer because n is nonsquare. From (32), and from

$$p_N < q_N\sqrt{n} + \frac{1}{N},$$

we obtain

$$\begin{aligned} |k_N| &= |p_N - q_N\sqrt{n}| (p_N + q_N\sqrt{n}) \\ &< \frac{1}{N} \left(2N\sqrt{n} + \frac{1}{N} \right) \\ &\leq 2\sqrt{n} + 1. \end{aligned}$$

Thus the nonzero integers k_N range over a finite set. Some fixed nonzero integer k therefore occurs for infinitely many distinct pairs (p, q) . Discarding finitely many terms if necessary, we may assume $p, q > 0$.

Among these infinitely many pairs with $p^2 - nq^2 = k$, two distinct pairs, say (p_1, q_1) and (p_2, q_2) , are congruent coordinatewise modulo $|k|$. Reorder them so that

$$p_1 + q_1\sqrt{n} > p_2 + q_2\sqrt{n}.$$

Define

$$(33) \quad \eta = \frac{(p_1 + q_1\sqrt{n})(p_2 - q_2\sqrt{n})}{k} = P + Q\sqrt{n}.$$

The coefficients are integers. Indeed,

$$P = \frac{p_1p_2 - nq_1q_2}{k}, \quad Q = \frac{q_1p_2 - p_1q_2}{k},$$

and the two numerators are congruent modulo $|k|$ to

$$p_2^2 - nq_2^2 = k \quad \text{and} \quad q_2p_2 - p_2q_2 = 0,$$

respectively. Furthermore,

$$N(\eta) = \frac{(p_1^2 - nq_1^2)(p_2^2 - nq_2^2)}{k^2} = 1.$$

Since $p_2 - q_2\sqrt{n} = k/(p_2 + q_2\sqrt{n})$, formula (33) also gives

$$\eta = \frac{p_1 + q_1\sqrt{n}}{p_2 + q_2\sqrt{n}} > 1.$$

Its conjugate is $\eta^{-1} \in (0, 1)$, and therefore

$$P = \frac{\eta + \eta^{-1}}{2} > 0, \quad Q = \frac{\eta - \eta^{-1}}{2\sqrt{n}} > 0.$$

Thus P, Q are positive integers satisfying $P^2 - nQ^2 = 1$. Renaming them p, q proves the lemma. $\square$

Lemma 4.2 (A parity-preserving unit). *Let $n \geq 2$ be nonsquare, and let $p + q\sqrt{n} > 1$ be any integer solution of $p^2 - nq^2 = 1$. If*

$$\lambda = (p + q\sqrt{n})^6 = P + Q\sqrt{n},$$

then

$$P \equiv 1 \pmod{2}, \quad Q \equiv 0 \pmod{2}.$$

Consequently, multiplication by every integral power of λ preserves both congruence conditions

$$U \equiv n^2 \pmod{2}, \quad V \equiv 0 \pmod{2}$$

on coefficient pairs $U + V\sqrt{n}$.

Proof. Multiplication by $p + q\sqrt{n}$ on the coefficient column vector $(U, V)^\top$ is represented by

$$M = \begin{pmatrix} p & nq \\ q & p \end{pmatrix}.$$

Its determinant is $p^2 - nq^2 = 1$, so its reduction modulo 2 lies in the group $\text{GL}_2(\mathbb{F}_2)$. This group has

$$(2^2 - 1)(2^2 - 2) = 3 \cdot 2 = 6$$

elements: there are three choices for a nonzero first column and then two choices for a second column outside its span. By Lagrange's theorem, the order of the residue class of M divides 6, and hence

$$M^6 \equiv I \pmod{2}.$$

On the other hand, multiplication by

$$(p + q\sqrt{n})^6 = P + Q\sqrt{n}$$

is represented by

$$M^6 = \begin{pmatrix} P & nQ \\ Q & P \end{pmatrix}.$$

Comparing the lower-left and diagonal entries modulo 2 gives $Q \equiv 0 \pmod{2}$ and $P \equiv 1 \pmod{2}$.

If $U' = PU + nQV$ and $V' = QU + PV$, then these congruences imply

$$U' \equiv U \pmod{2}, \quad V' \equiv V \pmod{2}.$$

The inverse unit is $\lambda^{-1} = P - Q\sqrt{n}$, which has the same parity properties, so the conclusion holds for every positive or negative integral power. $\square$

We next show that every parity-admissible solution of the generalized Pell equation has a unique representative in the finite window used in (8).

Proposition 4.3 (Finite seed reduction). *Let $n \geq 2$ be nonsquare, let $D_n = n^4 - 4$, and let $\lambda_n = P_n + Q_n\sqrt{n} > 1$ be as in (6). Then the set $\mathcal{C}_n$ in (8) is finite. More precisely, every $(A, B) \in \mathcal{C}_n$ satisfies*

$$(34) \quad |A| \leq \sqrt{D_n \lambda_n}, \quad |B| \leq \sqrt{\frac{D_n \lambda_n}{n}}.$$

Every pair $(U, V) \in \mathbb{Z}^2$ satisfying

$$(35) \quad U^2 - nV^2 = D_n, \quad U \equiv n^2 \pmod{2}, \quad V \equiv 0 \pmod{2}$$

has a unique representation

$$(36) \quad U + V\sqrt{n} = \varepsilon(A + B\sqrt{n})\lambda_n^m$$

with

$$\varepsilon \in \{\pm 1\}, \quad (A, B) \in \mathcal{C}_n, \quad m \in \mathbb{Z}.$$

Conversely, every expression on the right-hand side of (36) has integral coefficients satisfying (35).

Proof. Set

$$L = \sqrt{\frac{D_n}{\lambda_n}}.$$

Then the defining window for $\mathcal{C}_n$ is

$$(37) \quad L \leq \beta < L\lambda_n, \quad \beta = A + B\sqrt{n}.$$

Because $N(\beta) = D_n > 0$, its conjugate is

$$\bar{\beta} = A - B\sqrt{n} = \frac{D_n}{\beta} > 0.$$

From (37),

$$L < \bar{\beta} \leq L\lambda_n.$$

Thus both β and $\bar{\beta}$ are positive and at most

$$L\lambda_n = \sqrt{D_n \lambda_n}.$$

Since

$$A = \frac{\beta + \bar{\beta}}{2}, \quad B = \frac{\beta - \bar{\beta}}{2\sqrt{n}},$$

we obtain (34). Only finitely many integer pairs lie in these bounds, so $\mathcal{C}_n$ is finite.

Now let $\alpha = U + V\sqrt{n}$ satisfy (35). Since

$$\alpha\bar{\alpha} = D_n > 0,$$

the two real numbers α and $\bar{\alpha}$ have the same sign. There is therefore a unique $\varepsilon \in \{\pm 1\}$ for which

$$\gamma = \varepsilon\alpha > 0.$$

The half-open intervals

$$[L\lambda_n^m, L\lambda_n^{m+1}), \quad m \in \mathbb{Z},$$

partition the positive real axis. Hence there is a unique $m \in \mathbb{Z}$ such that

$$L\lambda_n^m \leq \gamma < L\lambda_n^{m+1}.$$

Put

$$\beta = \gamma \lambda_n^{-m}.$$

Then $L \leq \beta < L\lambda_n$. Because

$$\lambda_n^{-1} = P_n - Q_n \sqrt{n} \in \mathbb{Z}[\sqrt{n}],$$

the element β has integral coefficients, say $\beta = A + B\sqrt{n}$. Its norm is still D_n . By lemma 4.2, multiplication by λ_n^{-m} preserves the parity conditions in (35), so $(A, B) \in \mathcal{C}_n$. This proves existence of (36).

The sign ε is forced by the sign of α , the integer m is forced by the unique interval containing γ , and then $A + B\sqrt{n} = \gamma \lambda_n^{-m}$ is forced. Hence the representation is unique.

Conversely, take $(A, B) \in \mathcal{C}_n$, $m \in \mathbb{Z}$, and $\varepsilon \in \{\pm 1\}$. Both λ_n^m and λ_n^{-m} lie in $\mathbb{Z}[\sqrt{n}]$, so the product in (36) has integral coefficients. Norms multiply, and $N(\lambda_n) = 1$, so its norm is D_n . The parity conditions are preserved by lemma 4.2, and multiplication by -1 does not alter a congruence modulo 2. Thus (35) holds. $\square$

Proof of Theorem 1.1. Cases (I) and (II) are exactly propositions 3.1 and 3.3. It remains to prove Case (III).

Let $x = -n$, where $n \geq 2$ is nonsquare. By lemma 2.2, the map

$$(x, y, z) \mapsto U + V\sqrt{n} = (2y + n^2) + 2z\sqrt{n}$$

is a bijection from solutions with this fixed x to the coefficient pairs satisfying (35). By lemma 4.1, a positive norm-1 unit exists, so λ_n and $\mathcal{C}_n$ are defined. Proposition 4.3 gives a unique expression

$$U + V\sqrt{n} = \varepsilon(A + B\sqrt{n})\lambda_n^m$$

with $(A, B) \in \mathcal{C}_n$, $m \in \mathbb{Z}$, and $\varepsilon \in \{\pm 1\}$. Applying the inverse formulas (20) gives exactly (14). Conversely, every expression in (13) is parity-admissible by proposition 4.3, and hence gives a solution by lemma 2.2. This proves the classification and its uniqueness.

The finiteness of $\mathcal{C}_n$ is part of proposition 4.3. If $\mathcal{C}_n$ is empty, there are no solutions with $x = -n$. If it is nonempty, fix one seed $\beta = A + B\sqrt{n}$. The elements

$$\beta \lambda_n^m, \quad m \in \mathbb{Z},$$

are pairwise distinct because $\beta \neq 0$ and $\lambda_n > 1$. The bijection in lemma 2.2 then yields infinitely many distinct triples. Thus the fiber is either empty or infinite. $\square$

The orbit description can be written as an entirely integral recurrence.

Corollary 4.4 (Orbit recurrence). *Let $n \geq 2$ be nonsquare and write*

$$\lambda_n = P_n + Q_n \sqrt{n}.$$

If (U_m, V_m) is one admissible orbit in (13), then

$$(38) \quad \begin{aligned} U_{m+1} &= P_n U_m + n Q_n V_m, \\ V_{m+1} &= Q_n U_m + P_n V_m. \end{aligned}$$

In the original variables, the corresponding recurrence is

$$(39) \quad \begin{aligned} y_{m+1} &= P_n y_m + n Q_n z_m + \frac{(P_n - 1)n^2}{2}, \\ z_{m+1} &= Q_n y_m + P_n z_m + \frac{Q_n n^2}{2}. \end{aligned}$$

All coefficients and constant terms in (39) are integers.

Proof. Multiplying $U_m + V_m \sqrt{n}$ by $P_n + Q_n \sqrt{n}$ gives (38). Substituting

$$U_m = 2y_m + n^2, \quad V_m = 2z_m$$

and then solving for y_{m+1}, z_{m+1} gives (39). The parity relations $P_n \equiv 1 \pmod{2}$ and $Q_n \equiv 0 \pmod{2}$ from (7) make the two displayed halves integral. $\square$

5. AN EXPLICIT INFINITE ORBIT

The preceding theorem proves that any single solution with a fixed negative nonsquare value of x generates infinitely many. We finish with a concrete orbit.

Proposition 5.1. *Set*

$$y_0 = 9263, \quad z_0 = 1391,$$

and define recursively, for $m \geq 0$,

$$(40) \quad \begin{aligned} y_{m+1} &= 289y_m + 3480z_m + 3\,027\,600, \\ z_{m+1} &= 24y_m + 289z_m + 252\,300. \end{aligned}$$

Then, for every $m \geq 0$,

$$(41) \quad (-145, y_m, \pm z_m)$$

solves (1). The triples are pairwise distinct. Their images under the second symmetry in (15), namely

$$(42) \quad (-145, -21025 - y_m, \pm z_m),$$

are solutions as well.

Proof. A direct calculation gives

$$9263^2 + 145^2 \cdot 9263 - 145 \cdot 1391^2 + 1 = 0,$$

so $(-145, 9263, 1391)$ is a solution. In norm-form coordinates,

$$U_0 = 2y_0 + 145^2 = 39551, \quad V_0 = 2z_0 = 2782,$$

and therefore

$$U_0^2 - 145V_0^2 = 145^4 - 4.$$

Moreover,

$$289^2 - 145 \cdot 24^2 = 1.$$

The unit $289 + 24\sqrt{145}$ already has odd first coefficient and even second coefficient, so multiplication by it preserves the required parity conditions; taking a sixth power is unnecessary for this particular orbit. The general recurrence (39), with

$$n = 145, \quad P = 289, \quad Q = 24,$$

is exactly (40), because

$$145 \cdot 24 = 3480, \quad \frac{(289 - 1)145^2}{2} = 3\,027\,600, \quad \frac{24 \cdot 145^2}{2} = 252\,300.$$

Induction using the norm-preserving multiplication now proves that every triple in (41) is a solution.

The associated positive real numbers

$$U_m + V_m\sqrt{145} = (U_0 + V_0\sqrt{145})(289 + 24\sqrt{145})^m$$

are strictly increasing, because both factors are positive and $289 + 24\sqrt{145} > 1$. Hence the coefficient pairs, and therefore the triples, are pairwise distinct. Finally, (42) follows from the symmetry $y \mapsto -x^2 - y = -21025 - y$ in lemma 2.1. $\square$

The first new member of the positive- y branch is

$$(-145, 10\,545\,287, 876\,611).$$

Thus (1) has infinitely many integer solutions even after fixing $x = -145$. Together with theorem 1.1, this completes the unconditional and effective description of all integer solutions.